

From a Hartree–Fock reference to a VQE-UCCSD calculation

A hybrid quantum–classical workflow for electronic ground states

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Goal: the electronic ground-state energy

We want to approximate the lowest eigenvalue of an electronic Hamiltonian,

$$E_0 = \min_{|\psi\rangle} \frac{\langle\psi|\hat{H}|\psi\rangle}{\langle\psi|\psi\rangle}.$$

Exact diagonalisation scales as $\mathcal{O}(d^3)$ with Hilbert-space dimension $d = 2^{N_q}$ — intractable for moderate molecules.

The **Variational Quantum Eigensolver (VQE)** restricts the wave function to a parameterised ansatz,

$$|\psi(\theta)\rangle = U(\theta) |\Phi_0\rangle,$$

and minimises

$$E(\theta) = \langle\psi(\theta)|\hat{H}|\psi(\theta)\rangle \geq E_0.$$

For UCCSD the reference state is the Hartree–Fock determinant, $|\Phi_0\rangle = |\Phi_{\text{HF}}\rangle$. The rest of these notes develop *why* this is a good choice and *how* to realise it on a quantum computer.

Why the initial state matters

The ansatz is $|\Psi(\theta)\rangle = U(\theta) |\Phi_0\rangle$. The reference state $|\Phi_0\rangle$ controls:

- the region of Hilbert space reachable at a given circuit depth,
- the number of variational parameters needed for chemical accuracy,
- susceptibility to barren plateaus and local minima.

A bad reference (e.g. the qubit vacuum $|0 \cdots 0\rangle$) forces $U(\theta)$ first to build in the gross electronic structure before refining correlation.

A good reference encodes the *mean-field* physics already, so $U(\theta)$ only has to add electron correlation on top.

⇒ the Hartree–Fock state is the natural starting point inherited from classical quantum chemistry.

The electronic Hamiltonian in second quantisation

In second quantisation,

$$\hat{H} = \sum_{pq} h_{pq} \hat{a}_p^\dagger \hat{a}_q + \frac{1}{2} \sum_{pqrs} h_{pqrs} \hat{a}_p^\dagger \hat{a}_q^\dagger \hat{a}_s \hat{a}_r,$$

with one-body integrals

$$h_{pq} = \int d\mathbf{x} \phi_p^*(\mathbf{x}) \left(-\frac{1}{2} \nabla^2 - \sum_A \frac{Z_A}{|\mathbf{r} - \mathbf{R}_A|} \right) \phi_q(\mathbf{x}),$$

and two-body integrals

$$h_{pqrs} = \int d\mathbf{x}_1 d\mathbf{x}_2 \frac{\phi_p^*(\mathbf{x}_1) \phi_q^*(\mathbf{x}_2) \phi_r(\mathbf{x}_1) \phi_s(\mathbf{x}_2)}{|\mathbf{r}_1 - \mathbf{r}_2|}.$$

The single-particle basis $\{\phi_p\}$ is typically obtained from a preceding Hartree–Fock calculation, which is the next ingredient.

Hartree–Fock recap

The Hartree–Fock (HF) method approximates the many-electron wave function by a single Slater determinant,

$$|\Phi_{\text{HF}}\rangle = \frac{1}{\sqrt{N_e!}} \det[\varphi_{i_1}(x_1) \cdots \varphi_{i_{N_e}}(x_{N_e})],$$

with spin-orbitals $\{\varphi_p\}$ obtained self-consistently from the Roothaan–Hall equations.

In second quantisation this is just a product of creation operators acting on the Fock vacuum,

$$|\Phi_{\text{HF}}\rangle = \prod_{p \in \text{occ}} \hat{a}_p^\dagger |\text{vac}\rangle$$

where `occ` contains the N_e lowest-energy spin-orbitals.

HF captures the mean-field physics exactly. What it misses is the *correlation energy*,

$$E_{\text{corr}} = E_{\text{exact}} - E_{\text{HF}}.$$

Recovering E_{corr} on a quantum computer is precisely the job of $U(\theta)$.

From fermions to qubits

A qubit can encode the occupation of one spin-orbital,

$$|0\rangle_p \leftrightarrow \text{empty}, \quad |1\rangle_p \leftrightarrow \text{occupied}.$$

We need a mapping that respects fermionic antisymmetry. The fermionic Hamiltonian becomes

$$\hat{H} \longrightarrow \hat{H}_q = \sum_{\alpha} c_{\alpha} P_{\alpha}, \quad P_{\alpha} = \sigma_1^{(\alpha)} \otimes \sigma_2^{(\alpha)} \otimes \cdots \otimes \sigma_n^{(\alpha)},$$

a weighted sum of Pauli strings. Common mappings:

- **Jordan–Wigner (JW):** simple, local in occupation, non-local parity strings.
- **Parity:** swaps the locality trade-off, allows a two-qubit reduction via $\mathbb{Z}_2$ symmetries.
- **Bravyi–Kitaev (BK):** $\mathcal{O}(\log N)$ locality in both occupation and parity.

Under Jordan–Wigner,

$$\hat{a}_p^{\dagger} = \left(\prod_{q < p} Z_q \right) \frac{X_p - iY_p}{2}.$$

The VQE objective then reads $E(\boldsymbol{\theta}) = \sum_{\alpha} c_{\alpha} \langle P_{\alpha} \rangle_{\boldsymbol{\theta}}$.

The HF state under Jordan–Wigner

With JW ordering (spin-orbitals indexed $0, 1, \dots, M - 1$, occupied first),

$$|\Phi_{\text{HF}}\rangle_{\text{JW}} = \underbrace{|1\rangle |1\rangle \cdots |1\rangle}_{N_e \text{ occupied}} \underbrace{|0\rangle |0\rangle \cdots |0\rangle}_{M - N_e \text{ virtual}}.$$

This is a product state! It is prepared from $|0\rangle^{\otimes M}$ by applying X gates on the first N_e qubits.

For H_2 in the minimal STO-3G basis,

$$M = 4 \text{ spin-orbitals}, \quad N_e = 2 \implies |\Phi_{\text{HF}}\rangle = |1100\rangle.$$

Different mappings produce different bit patterns for the *same* physical state:

Mapping	H_2 HF state	# qubits
Jordan–Wigner	$ 1100\rangle$	4
Parity	$ 0110\rangle$	4
Parity + $\mathbb{Z}_2$	$ 01\rangle$	2
Bravyi–Kitaev	$ 1110\rangle$	4

Never hard-code the bit pattern — derive it from the mapper.

A bare-bones HF initialiser in Qiskit

Under Jordan–Wigner, preparing $|\Phi_{\text{HF}}\rangle$ takes N_e Pauli- X gates:

```
from qiskit import QuantumCircuit, QuantumRegister

def hartree_fock_jw(n_spin_orbitals, n_electrons):
    Prepare a Hartree-Fock reference state under JW mapping.
    Occupied orbitals are the lowest n_electrons indices.
    qr = QuantumRegister(n_spin_orbitals, name=q)
    qc = QuantumCircuit(qr, name=HF)
    for i in range(n_electrons):
        qc.x(qr[i]) # flip |0> -> |1> on occupied orbitals
    return qc

hf_circuit = hartree_fock_jw(n_spin_orbitals=4, n_electrons=2)
print(hf_circuit)
```

Schematic output:

```
q_0: -X-
q_1: -X-
q_2: ---
q_3: ---
```

That is the entire HF state preparation under JW.

Using Qiskit Nature's built-in HF initial state

For production work, let the library handle JW, parity (with two-qubit reduction), and BK consistently:

```
from qiskit_nature.second_q.circuit.library import HartreeFock
from qiskit_nature.second_q.mappers import JordanWignerMapper

mapper = JordanWignerMapper()

hf_init = HartreeFock(
    num_spatial_orbitals = 2, # H2 in STO-3G has 2 spatial orbitals
    num_particles = (1, 1), # (alpha, beta) electrons
    qubit_mapper = mapper,
)
print(hf_init.draw())
```

Switching mapper = switching the bit-pattern that represents HF; the library does the bookkeeping. With $|\Phi_{\text{HF}}\rangle$ in hand, we can now ask the quantum computer to add the correlation.

Classical coupled cluster

In classical coupled-cluster (CC) theory, the correlated wave function is

$$|\psi_{\text{CC}}\rangle = e^{\hat{T}} |\Phi_{\text{HF}}\rangle, \quad \hat{T} = \hat{T}_1 + \hat{T}_2 + \hat{T}_3 + \dots$$

with single and double excitation operators

$$\hat{T}_1 = \sum_{ia} t_i^a \hat{a}_a^\dagger \hat{a}_i, \quad \hat{T}_2 = \frac{1}{4} \sum_{ijab} t_{ij}^{ab} \hat{a}_a^\dagger \hat{a}_b^\dagger \hat{a}_j \hat{a}_i.$$

Indices: i, j are occupied spin-orbitals; a, b are virtual spin-orbitals.

Each excitation promotes one (or two) electrons from below to above the Fermi level of the HF reference. The coefficients t_i^a , t_{ij}^{ab} are the cluster amplitudes that encode electron correlation.

Problem for quantum hardware: the operator $e^{\hat{T}}$ is *not* unitary, so it cannot be implemented directly as a quantum circuit.

Unitary coupled cluster

The fix is to use an anti-Hermitian generator. Define

$$\hat{A} = \hat{T} - \hat{T}^\dagger, \quad \hat{A}^\dagger = -\hat{A}.$$

Then $e^{\hat{A}}$ is unitary, and the unitary coupled-cluster ansatz reads

$$|\Psi_{\text{UCC}}\rangle = e^{\hat{T} - \hat{T}^\dagger} |\Phi_{\text{HF}}\rangle.$$

Truncating at singles and doubles gives **UCCSD**,

$$|\Psi_{\text{UCCSD}}\rangle = e^{\hat{T}_1 + \hat{T}_2 - \hat{T}_1^\dagger - \hat{T}_2^\dagger} |\Phi_{\text{HF}}\rangle.$$

Two crucial properties that justify starting from the HF reference:

- At $\theta = 0$ all amplitudes vanish and $|\Psi(0)\rangle = |\Phi_{\text{HF}}\rangle$. The optimiser starts at the (known) HF energy — a sensible baseline.
- Each excitation operator promotes electrons from occupied to virtual orbitals, i.e. it generates correlation *on top of* HF.

Single and double excitation operators

Single excitation $i \rightarrow a$ has anti-Hermitian generator

$$\hat{\tau}_i^a = \hat{a}_a^\dagger \hat{a}_i - \hat{a}_i^\dagger \hat{a}_a,$$

giving the unitary

$$U_i^a(\theta_i^a) = \exp[\theta_i^a (\hat{a}_a^\dagger \hat{a}_i - \hat{a}_i^\dagger \hat{a}_a)].$$

This rotates between the reference and singly excited determinants.

Double excitation $ij \rightarrow ab$ has generator

$$\hat{\tau}_{ij}^{ab} = \hat{a}_a^\dagger \hat{a}_b^\dagger \hat{a}_j \hat{a}_i - \hat{a}_i^\dagger \hat{a}_j^\dagger \hat{a}_b \hat{a}_a,$$

and unitary

$$U_{ij}^{ab}(\theta_{ij}^{ab}) = \exp[\theta_{ij}^{ab} \hat{\tau}_{ij}^{ab}].$$

Doubles are essential for capturing dynamical electron correlation.

Trotterised UCCSD ansatz

The exact UCCSD unitary is

$$U_{\text{UCCSD}}(\boldsymbol{\theta}) = \exp \left[\sum_{ia} \theta_i^a \hat{\tau}_i^a + \sum_{ijab} \theta_{ij}^{ab} \hat{\tau}_{ij}^{ab} \right].$$

Excitation generators generally do not commute, $[\hat{\tau}_\mu, \hat{\tau}_\nu] \neq 0$, so a single exponential cannot be implemented exactly. In practice one uses a first-order Trotter approximation,

$$U_{\text{UCCSD}}(\boldsymbol{\theta}) \approx \prod_{ia} e^{\theta_i^a \hat{\tau}_i^a} \prod_{ijab} e^{\theta_{ij}^{ab} \hat{\tau}_{ij}^{ab}}.$$

Each factor is a Pauli-string rotation that compiles to a short sequence of CNOT and single-qubit gates.

The Trotter error is absorbed by the variational optimisation: the parameters $\boldsymbol{\theta}$ are re-tuned to minimise the energy of whatever circuit is actually executed.

Quantum circuit construction

The full UCCSD trial state on a quantum computer is

$$|\Psi(\boldsymbol{\theta})\rangle = U_{\text{UCCSD}}(\boldsymbol{\theta}) |\Phi_{\text{HF}}\rangle.$$

Workflow:

- 1 Prepare $|\Phi_{\text{HF}}\rangle$ on the qubits (N_e Pauli- X gates under JW).
- 2 Apply all selected single-excitation unitaries $\prod_{ia} e^{\theta_i^a \hat{\tau}_i^a}$.
- 3 Apply all selected double-excitation unitaries $\prod_{ijab} e^{\theta_{ij}^{ab} \hat{\tau}_{ij}^{ab}}$.
- 4 Measure expectation values of the Pauli strings in $\hat{H}_q$.

The HF preparation in step 1 is essentially free (a single layer of X gates). Steps 2–3 dominate the depth: UCCSD circuits contain many excitations and become deep for larger systems.

Energy measurement

After fermion-to-qubit mapping the Hamiltonian is

$$\hat{H}_q = \sum_{\alpha} c_{\alpha} P_{\alpha}.$$

For a fixed parameter vector θ we estimate

$$E(\theta) = \sum_{\alpha} c_{\alpha} \langle \Psi(\theta) | P_{\alpha} | \Psi(\theta) \rangle.$$

Each Pauli string is measured separately on the quantum device. For example,

$$P_{\alpha} = Z_0 Z_1 X_2 Y_3,$$

requires rotating qubits 2 and 3 into the X and Y bases (Hadamard, $S^{\dagger}H$) before a Z -basis measurement.

The number of distinct Pauli strings grows polynomially with system size, and grouping commuting terms reduces the number of measurement circuits.

The hybrid quantum–classical loop

VQE is hybrid: each iteration combines quantum estimation with classical optimisation.

$$\boldsymbol{\theta}^{(k)} \longrightarrow \text{quantum circuit} \longrightarrow E(\boldsymbol{\theta}^{(k)}) \longrightarrow \text{classical optimiser}.$$

The optimiser updates the parameters,

$$\boldsymbol{\theta}^{(k+1)} = \boldsymbol{\theta}^{(k)} + \Delta\boldsymbol{\theta}^{(k)},$$

until convergence,

$$|E^{(k+1)} - E^{(k)}| < \epsilon.$$

The variational principle guarantees

$$E_{\text{VQE}} = \min_{\boldsymbol{\theta}} E(\boldsymbol{\theta}) \geq E_0,$$

so improving the ansatz (more excitations, deeper Trotter, larger basis) can only lower the variational energy.

Gradients and the parameter-shift rule

Gradient-free optimisers (COBYLA, SPSA, SLSQP) are common, but gradients can also be computed on hardware. For a parameter θ_μ ,

$$\frac{\partial E}{\partial \theta_\mu} = \langle \partial_\mu \Psi | \hat{H} | \Psi \rangle + \langle \Psi | \hat{H} | \partial_\mu \Psi \rangle.$$

For Pauli-generated rotations $U_\mu(\theta_\mu) = e^{-i\theta_\mu P_\mu/2}$, the **parameter-shift rule** gives an exact analytic gradient from two extra circuit evaluations:

$$\frac{\partial E}{\partial \theta_\mu} = \frac{1}{2} \left[E\left(\theta_\mu + \frac{\pi}{2}\right) - E\left(\theta_\mu - \frac{\pi}{2}\right) \right].$$

This is structurally similar to a finite difference but is *exact* for these gate generators — no step-size bias.

Algorithm: VQE-UCCSD with a Hartree–Fock reference

- 1 Choose a molecular geometry and a one-particle basis (e.g. STO-3G).
- 2 Compute one- and two-body integrals h_{pq} and h_{pqrs} (PySCF, Psi4, ...).
- 3 Assemble the second-quantised Hamiltonian

$$\hat{H} = \sum_{pq} h_{pq} \hat{a}_p^\dagger \hat{a}_q + \frac{1}{2} \sum_{pqrs} h_{pqrs} \hat{a}_p^\dagger \hat{a}_q^\dagger \hat{a}_s \hat{a}_r.$$

- 4 Map to a qubit Hamiltonian via JW / parity / BK: $\hat{H}_q = \sum_{\alpha} c_{\alpha} P_{\alpha}$.
- 5 Prepare the Hartree–Fock reference $|\Phi_{\text{HF}}\rangle$.
- 6 Build the UCCSD ansatz on top of it: $U(\theta) = e^{\hat{T}_1 + \hat{T}_2 - \hat{T}_1^\dagger - \hat{T}_2^\dagger}$.
- 7 Initialise $\theta = 0$ (so the trial state starts at $|\Phi_{\text{HF}}\rangle$).
- 8 Measure $E(\theta) = \langle \Psi(\theta) | \hat{H}_q | \Psi(\theta) \rangle$.
- 9 Update θ with a classical optimiser; repeat 8–9 until $|\Delta E| < \epsilon$.

Putting it all together: H₂ with Qiskit Nature

```
from qiskit_nature.second_q.drivers import PySCFDriver
from qiskit_nature.second_q.mappers import JordanWignerMapper
from qiskit_nature.second_q.circuit.library import HartreeFock, UCCSD
from qiskit_algorithms import VQE
from qiskit_algorithms.optimizers import SLSQP
from qiskit.primitives import Estimator
import numpy as np

# 1-3. Electronic structure problem and second-quantised Hamiltonian
driver = PySCFDriver(atom='H 0 0 0; H 0 0 0.735', basis='sto3g')
problem = driver.run()

# 4. Map fermions -> qubits
mapper = JordanWignerMapper()
qubit_op = mapper.map(problem.hamiltonian.second_q_op())

# 5. Hartree-Fock initial state
hf_state = HartreeFock(problem.num_spatial_orbitals,
                        problem.num_particles, mapper)

# 6. UCCSD ansatz built ON TOP OF the HF reference
ansatz = UCCSD(problem.num_spatial_orbitals,
               problem.num_particles, mapper,
               initial_state=hf_state)

# 7-9. Run VQE starting from theta = 0 (state = |Phi_HF>)
vqe = VQE(Estimator(), ansatz, SLSQP(maxiter=200),
```

What this code is doing, line by line

- **Driver + problem:** PySCF computes the HF orbitals and the integrals h_{pq} , h_{pqrs} in that basis.
- **Mapper:** converts $\hat{H}$ into a sum of Pauli strings $\hat{H}_q$.
- **HartreeFock:** a circuit that prepares $|\Phi_{\text{HF}}\rangle$ — under JW just N_e Pauli- X gates.
- **UCCSD(initial_state=hf_state):** the trial state is $U_{\text{UCCSD}}(\theta)|\Phi_{\text{HF}}\rangle$.
- **initial_point=zeros:** at iteration 0, $\theta = 0$, so $|\Psi(0)\rangle = |\Phi_{\text{HF}}\rangle$ and $E(0) = E_{\text{HF}}$.
- **VQE:** the hybrid loop runs — the optimiser only has to find the *correlation* energy $E_{\text{corr}} = E_{\text{exact}} - E_{\text{HF}}$, a small fraction of the total.

For $\text{H}_2/\text{STO-3G}$ this converges in a handful of iterations and reproduces the FCI energy to machine precision.

Scaling of UCCSD

Let

$$N_{\text{occ}} = \# \text{ occupied spin-orbitals}, \quad N_{\text{virt}} = \# \text{ virtual spin-orbitals}.$$

Number of excitation amplitudes:

$$N_{\text{singles}} \sim N_{\text{occ}} N_{\text{virt}}, \quad N_{\text{doubles}} \sim N_{\text{occ}}^2 N_{\text{virt}}^2.$$

Each amplitude contributes a Trotter factor $e^{\theta_\mu \hat{\tau}_\mu}$, which compiles to multiple CNOTs. The circuit depth therefore grows roughly as $\mathcal{O}(N_{\text{occ}}^2 N_{\text{virt}}^2)$, which becomes prohibitive for medium-sized molecules on near-term devices.

Mitigations: active-space selection, point-group/spin symmetries, k-UpCCGSD, ADAPT-VQE (grow the operator pool adaptively), tapering off $\mathbb{Z}_2$ symmetries (e.g. two-qubit reduction).

Why UCCSD is useful — and where it hurts

Strengths:

- Physically motivated; direct link to classical quantum chemistry.
- Captures dynamical electron correlation systematically.
- Unitary form is natural for quantum circuits.
- Size-extensive in the untruncated limit.

Limitations:

- Deep circuits and many variational parameters.
- Trotter errors (mitigated by re-optimising θ).
- Heavy measurement overhead — thousands of Pauli strings for larger molecules.
- Optimisation landscape can suffer from local minima and barren plateaus.
- HF is a poor reference for strongly correlated systems (stretched bonds, transition metals).

Beyond a single-determinant reference

When HF fails (strong static correlation), the same machinery still applies but with a richer $|\Phi_0\rangle$:

- **Multi-determinant references** (CASSCF, CISD): prepared by an entangling layer of controlled rotations on top of the HF X-gates.
- **Symmetry-broken (UHF) orbitals**: cheaper, but lose spin symmetry.
- **ADAPT-VQE**: grow the operator pool one excitation at a time, partly compensating for a poor reference.

The structure of the VQE-UCCSD workflow does not change — only the first circuit block (initial state) and possibly the operator pool. Everything else (mapping, measurement, optimisation, parameter-shift gradients) carries over unchanged.

Conceptual summary

The VQE-UCCSD method combines

quantum chemistry + unitary coupled cluster + quantum measurement + classical optimisation.

The central variational problem is

$$E_{\text{VQE}} = \min_{\boldsymbol{\theta}} \langle \Phi_{\text{HF}} | U_{\text{UCCSD}}^\dagger(\boldsymbol{\theta}) \hat{H} U_{\text{UCCSD}}(\boldsymbol{\theta}) | \Phi_{\text{HF}} \rangle.$$

Take-home messages:

- The Hartree–Fock state is the natural reference; under Jordan–Wigner it is a single product state $|1 \cdots 1 0 \cdots 0\rangle$.
- Starting VQE at $\boldsymbol{\theta} = 0$ means $|\Psi(0)\rangle = |\Phi_{\text{HF}}\rangle$ and $E(0) = E_{\text{HF}}$; only the correlation energy needs to be recovered.
- UCCSD adds correlation through anti-Hermitian single and double excitations, $e^{\hat{T} - \hat{T}^\dagger}$, which is unitary and circuit-implementable.
- The quantum computer prepares the correlated wave function; the classical computer updates the cluster amplitudes.
- For strongly correlated systems, replace HF with a multi-determinant or adaptive reference — the rest of the pipeline is unchanged.

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